

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460709.8804 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.057
Transit depth (Rp/Rs)^2	3.82 +/- 2.24 [%]
Orbital Inclination (inc)	86.35 +/- 2.86
Airmass coefficient 1 (a1)	0.2 +/- 0.11
Airmass coefficient 2 (a2)	1.09 +/- 0.94
Scatter in the residuals of the lightcurve fit is	49.68 %
Best Comparison Star	None
Optimal Aperture	2.41
Optimal Annulus	3.0
Transit Duration (day)	0.0394 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.195 ± 0.057 vs 0.1594 (deviation 0.4σ)
Duration fit vs published	0.0394 d vs 0.0512 d (deviation 1.95σ)
Mid-time O-C	-6.6 min
Depth SNR	2.2
Residual scatter	49.68 %

Light curve

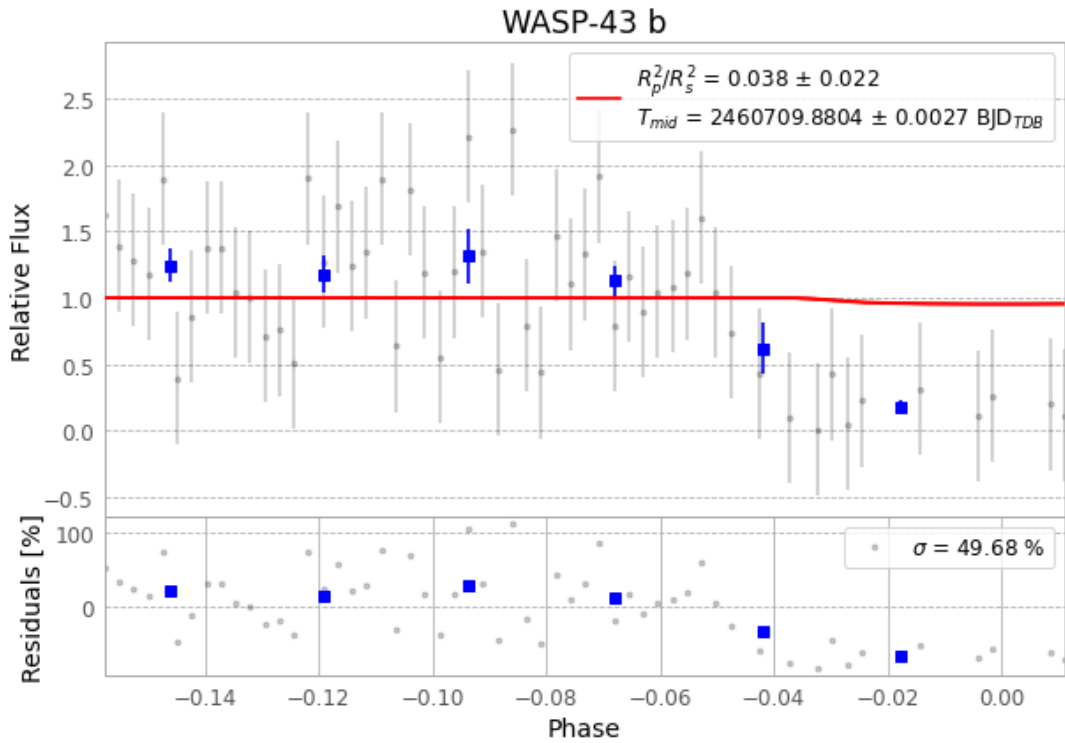


Figure 1 — FinalLightCurve_WASP-43 b_2025-02-03.png (SHA-256 42c2367ea186d2ec...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [241, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d0f687f5ca09074...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

69 FITS frames (45 MB), WASP-43250203055419.FITS → WASP-43250203091815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250203.log (5c302cca2d24...), FinalLightCurve_WASP-43 b_2025-02-03.png (42c2367ea186...), FinalLightCurve_WASP-43 b_2025-02-03.csv (e95817618d83...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 06:19Z.